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OW-DLN: A novel open-world decoupled learning network for fabric defect detection

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ABSTRACT

Fabric defect detection plays a critical role in textile manufacturing; however, conventional methods are inherently limited in recognizing unseen defects due to their reliance on predefined defect classes. To address this issue, we propose OW-DLN, an open-world fabric defect detection network that facilitates effective detection and incremental learning of unknown defects by unifying generative reconstruction and decoupled feature learning for known and unknown defects. The framework consists of three components. A Mask-Free Inpainting (MFI) module that reconstructs images with defects into defect-free image representations, thereby guiding semantic feature learning. A Dual-View Pseudo-label Generator (DVPG) module that generates more reliable pseudo-labels, reducing the omission of unknown samples and enabling the model to better capture features of unknown defects, while a Decoupled Incremental Learning (DIL) module that segregates the feature learning of known and unknown defects, effectively mitigating catastrophic forgetting. Extensive experiments on the Tianchi Benchmark and a self-collected fabric defect dataset demonstrate the superior performance of OW-DLN in open-world and incremental settings, achieving an unknown recall of 0.156 and 0.510 mAP@50 under class expansion settings, surpassing state-of-the-art (SOTA) OWOD methods and conventional fabric defect detection approaches.

1. Introduction

Defect detection plays a fundamental role in the fabric manufacturing process, and the application of computer vision techniques to automate fabric detection has attracted considerable interest. The wide variety of defect types, together with the high visual similarity among defect regions, substantially increases the difficulty and cost of reliable defect detection [1]. Traditional algorithms for fabric defect detection, such as thresholding, edge detection and texture analysis, generally exhibit limited robustness in real-world applications. These limitations stem from their sensitivity to variations in illumination, background noise, and the intrinsic texture complexity of textiles [2]. In recent years, substantial advancements have been made in the detection of fabric defects through the application of deep learning models, particularly convolutional neural networks (CNNs) and Transformer-based models, which utilize large-scale, labeled datasets [3]. However, these methods typically struggle to handle defect categories that are absent from the training data, exposing a fundamental limitation in their ability to generalize to previously unseen patterns. As a result, they frequently fail to identify new or unknown defects that may arise under dynamic production conditions [3].

In contrast to closed-world detection pipelines, open-world object detection approaches have been developed to enhance the generalization capability of detection models. Specifically, open-world detection methods introduce mechanisms, including unknown object awareness and incremental learning, to recognize novel and rare objects [4]. Despite the notable potential of open-world approaches in practical applications, their actual deployment for fabric defect detection still encounters the following challenges: (1) Fabrics have textured surfaces that yield low contrast between defects and the background, making reliable defect localization particularly challenging. As a result, during incrementally training on a small number of novel defects, a substantial fraction of defects cannot be automatically annotated. Consequently, open-world detection models may exhibit reduced recognition accuracy. (2) The visual features of different defect classes are not sufficiently distinct, resulting in confusion between unknown and known classes and exacerbating catastrophic forgetting.

In practical textile manufacturing, defect-type identification primarily serves quality inspection for remediation and product screening. The continual emergence of new defects and the difficulty of exhaustive data annotation motivate an open-world formulation that can

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proactively discover previously unseen defect types and incrementally learn them without degrading performance on known defects. To address these challenge, we introduce Open-world Decoupled Learning Network (OW-DLN), a novel framework designed to address these challenges. This research explicitly characterizes fabric defect detection as an Open-world Object Detection (OWOD) problem. The objectives are twofold: (i) to identify previously unrecognized defects during testing, and (ii) to incrementally learn these novel categories while maintaining detection accuracy for previously identified categories. To fulfill these goals, OW-DLN incorporates four synergistic modules: a Mask-Free Inpainting (MFI) module that discerns unknown cues through reconstruction residuals, a Dual-View Pseudo-label Generator (DVPG) that generates dependable pseudo-labels for unknown samples, a Decoupled Incremental Learning (DIL) module that alleviates catastrophic forgetting during the expansion of classes, where incremental learning denotes the ongoing adaptation of the detector to new defect categories without necessitating retraining from the outset or forgetting known ones, thus ensuring consistent performance in dynamic industrial settings, and a Defect-Aware Attention Module (DAAM) that enhances defect-focused representation learning. The primary contributions of this study are as follows:

1. A generative model-driven scheme is proposed, which aims to generate realistic defect-free images to enhance the pattern diversity and mode coverage for pseudo-label during unknown-class annotation.
2. A decoupled incremental learning strategy and network structure are introduced to separate the representation learning of known and unknown defect classes, thereby mitigating catastrophic forgetting during the expansion of classes.
3. We propose OW-DLN, a unified framework for open-world fabric defect detection that achieves state-of-the-art performance under both open-world and conventional closed-world settings. Specifically, OW-DLN achieves 0.153 Unknown Recall and 0.680 mAP@50 in the open-world detection task, and 0.502 mAP@50 in the all-classes setting.

2. Related work

2.1. Fabric defect detection

Neural networks, and particularly deep learning models, have revolutionized defect detection across various industries. Traditional image processing techniques often perform poorly under background noise, textural variations, and inconsistent illumination. To surmount these limitations, researchers have increasingly turned to deep learning for automated defect identification, with deep learning methods demonstrating considerable success in fabric defect detection in recent years [5].

Convolutional Neural Networks (CNNs) are widely used for defect detection due to their strong feature extraction capabilities. These networks autonomously learn hierarchical representations, making them effective at identifying complex patterns and anomalies in images. Hu et al. [6] introduced a spatiotemporally resolved method (TST-F) for measuring fabric mechanical properties and predicting touch sensations. Lin et al. [7] developed Fabric4show, a real-time vision system that integrates Faster R-CNN for defect detection and localization, reducing computational resource requirements while maintaining high accuracy. Hassan et al. [8] proposed an enhanced deep convolutional neural network (DCNN) for fabric defect detection, and Liu et al. [9] introduced GC-Net, a steel surface defect detection model with a global attention module, multi-scale feature fusion, and data augmentation to address data scarcity. Yan et al. [10] explored the use of micro 3D textured surfaces in electronic products to enhance appearance and user experience.

YOLO models are renowned for their real-time object detection capabilities, making them suitable for defect detection in high-speed industrial environments. The architecture of YOLO divides the input image into a grid and predicts bounding boxes and class probabilities for each grid cell, enabling efficient and accurate detection. Wang et al. [11] improved the YOLOv5 algorithm for fabric defect detection by integrating ASFF for better multi-scale feature fusion and transformer mechanisms to enhance feature focus. Hu et al. [12] proposed an efficient, lightweight deep convolutional neural network-based algorithm to enhance defect detection performance while maintaining inference speed. Zhou et al. [13] proposed MPA-YOLO to improve feature representation and defect localization. Despite these advancements in fabric defect detection methods using detection models, challenges remain. For example, detection models require the collection of large amounts of data and simultaneous manual annotation. However, collecting large-scale labeled datasets remains challenging due to the diversity of fabric defects, which arises from the wide range of defect morphologies, subtle texture inconsistencies, varied illumination responses, and the high visual similarity between normal and defective regions [14].

Generative models have demonstrated significant potential in unsupervised anomaly and defect detection by modeling the distribution of normal data. In the context of fabric defect detection, these models are trained to reconstruct defect-free images, and any discrepancies in the reconstruction process can indicate the presence of defects. Niu et al. [15] proposed a novel defect attention module in the discriminator of Cycle-GAN to better eliminate defects with weak signals. Zhang et al. [16] proposed an attention-based vector quantisation variational autoencoder (AVQ-VAE) for color-patterned fabric defect detection. Lee et al. [17] proposed a fabric inspection network (FIN) that combines variational autoencoder (VAE) and generative adversarial network (GAN) for unsupervised fabric defect detection. Zhang et al. [18] proposed a weakly supervised method, CADN, which performs defect classification and localization using only image-level tags, and employs knowledge distillation to balance accuracy and real-time performance.

2.2. Open-world object detection

Open-world object detection is an emerging research area that focuses on recognizing and learning from objects not present in initial training sets. This field incrementally expands its knowledge base as new class labels are introduced, thereby addressing the limitations of traditional closed-set models that assume all possible classes are known beforehand.

Joseph et al. [4] introduced OWOD and ORE, object detection framework that identifies unseen objects as "unknown" and incrementally learns new categories using contrastive clustering and energy-based identification. Gupta et al. [19] proposed OW-DETR, a transformer-based model for detecting unknown objects and separating them with pseudo-labeling, novelty classification, and objectness scoring. Cheng et al. [20] introduced YOLO-World, leveraging RepVL-PAN and region-text contrastive loss for open-vocabulary detection. Zheng et al. [21] proposed CDAL, which uses counterfactual reasoning to reduce spurious correlations in open-world model attribution. Guo et al. [22] explored robust semi-supervised learning with incomplete labels, and Yang et al. [23] developed POSE to progressively separate known and unknown distributions for better attribution.

However, despite significant advancements in open-world object detection, its application to fabric defect detection remains at a nascent stage. Fabric defect detection frequently encounters challenges such as diverse categories, complex defect types, and significant background variations. Traditional closed-set models exhibit limited capabilities in addressing new types of defects. The open-world detection model addresses this shortfall effectively by labeling unseen defect types as "unknown" and incrementally learning about new defect types. By adopting the open-world detection approach, the fabric defect detection system can autonomously identify and classify new defect types during the production process without relying on a fixed set of defect labels, thereby significantly enhancing the system's adaptability and accuracy.

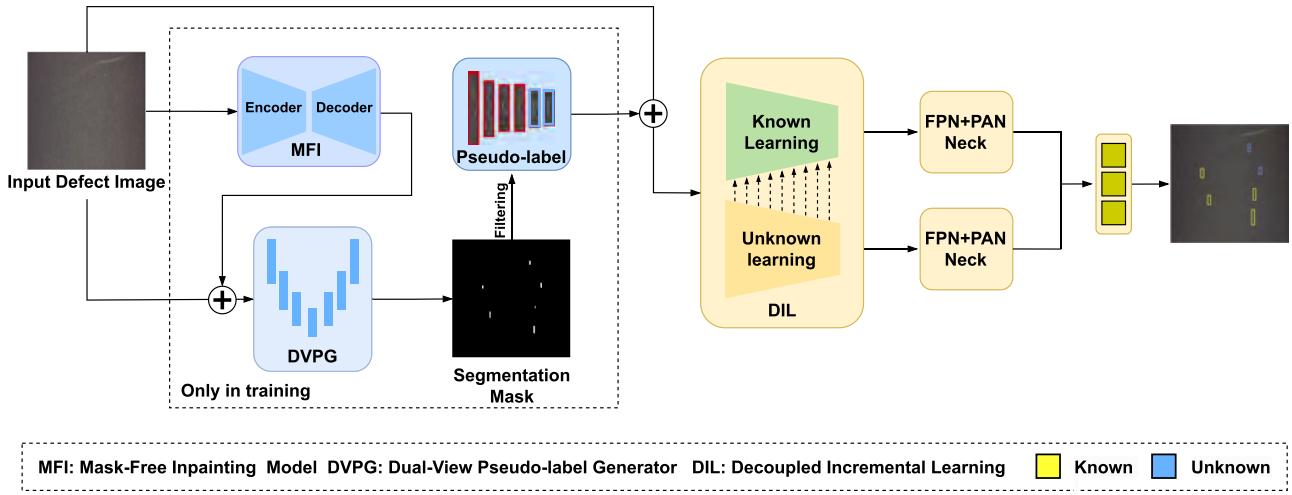

Fig. 1. Framework of OW-DLN. Mask-Free Inpainting (MFI): Reconstructing pseudo defect-free images; Dual-View Pseudo-label Generator (DVPG): Generating more reliable pixel-level pseudo labels; Decoupled Incremental Learning (DIL): Separates the feature learning of known and unknown defects.

3. Proposed method

Our approach comprises three components, Mask-Free Inpainting (MFI), Dual-View Pseudo-label Generator (DVPG), and Decoupled Incremental Learning (DIL), as shown in Fig. 1. Within this framework, MFI reconstructs defect-free fabric representations to accentuate potential anomalies, meanwhile DVPG generates more reliable pseudo-labels, and DIL facilitates the model's incremental learning of the unknown defect classes while preserving knowledge of known categories. The details of each component are discussed in the subsequent subsections.

3.1. Mask-free inpainting (MFI)

In open-world detection, assigning pseudo-labels to unknown class targets is key for incremental learning. We propose a mask-free inpainting model that reconstructs defect-free images and exploits the residuals between the reconstructed and original images to identify unknown defects. Unlike traditional inpainting methods, our model does not require paired data or explicit masks. During inference, the absence of defect location information leads to dense sliding-window sampling with small masks, causing significant computational redundancy and reduced efficiency.

Accordingly, we introduce a two-stage, mask-free inpainting strategy. In the first stage, we train a prior model on defect-free samples by applying random masks to reconstruct the masked regions. In the second stage, we distill the learned priors into a lightweight autoencoder. The masking process can be effectively decoupled from the inpainting task, thereby enabling direct, mask-independent reconstruction as shown in Fig. 2.

We divide the dataset into two parts: the defect-free image dataset $\mathcal{D}_{df} = \left\{x_{i=1}^{df}\right\}$ and the defect image dataset $\mathcal{D}_d = \left\{x_{j=1}^{d}\right\}$. Here, x_i^{df} denotes the ith defect-free image, and x_j^d denotes the jth defective image. To increase the robustness of the prior model M_p, we perform random irregular masking on each x_i^{df}, resulting in a masked image $\tilde{x}_i^{df}$. The mask image $\tilde{x}_i^n$ is then fed into M_p to improve the tetxture consistency between the masked and unmasked regions, we use a pretrained latent Stable Diffusion v1.5 model as the prior model M_p in this stage as follows:

$$\hat{x}_i^{df} = \mathcal{D}\left(\frac{1}{\sqrt{\tilde{\alpha}_T}}\mathcal{E}(\tilde{x}_i^{df}) - \sqrt{\frac{1-\tilde{\alpha}_T}{\tilde{\alpha}_T}}\epsilon_\theta\left(\mathcal{E}(\tilde{x}_i^{df}), T, m_i\right)\right) \tag{1}$$

where $\mathcal{E}(\cdot)$ is the VQ-VAE encoder that projects inputs into the latent space, $\mathcal{D}(\cdot)$ is the VAE decoder that maps latent variables back to image space, and $\epsilon_\theta(\cdot)$ is the U-Net noise predictor conditioned on latent variables, timestep T, and mask m_i, $m_i \in \{0,1\}^{H\times W}$ being the binary mask for defect-free supervision, $\tilde{\alpha}_T = \prod_{t=1}^{T}\alpha_t$ is the cumulative product of diffusion scheduling coefficients. This process ensures that the prior model is exposed to a variety of defect-free reconstructions and learns high-quality texture priors under masked conditions.

By employing the inpainting model M_p, we can obtain the reconstructed defect image dataset $\hat{\mathcal{D}}_d = \left\{\hat{x}_{j=1}^{M}\right\}$ generated from the defect image dataset $\mathcal{D}_d = \left\{x_{j=1}^{M}\right\}$, and $\hat{\mathcal{D}}_d$ is subsequently used to train the mask-free inpainting model in the second stage.

To facilitate the generation of high-fidelity, defect-free images, we design a Mask-Free Inpainting (MFI) model, composed of an encoder-decoder architecture. The encoder (E_{MFI}) extracts compact latent representations from defective inputs, whereas the decoder (D_{MFI}) is trained to reconstruct defect-free images from these representations without relying on explicit masks. To enhance the semantic fidelity and structural alignment of the reconstructed images, we further introduce a cross-module feature distillation strategy, wherein multi-level representations from the decoder D_p of a prior model are transferred to the decoder (D_{MFI}) of our MFI model. This approach allows the MFI decoder to leverage the rich semantic context encoded by the prior Stable Diffusion model. Thus faster convergence can be achieved and defect-free textures can be more accurately reconstructed. Representative results are shown in Fig. 3.

Let $D_p^{(l)}(x_j^d) \in \mathbb{R}^{C_l\times H_l\times W_l}$ denote the lth layer feature map extracted from the D_p when given a defective input x_j^d, and let $D_{MFI}^{(l')}(x_j^d)$ represent the corresponding feature map from the decoder D_p. The cross-module knowledge distillation loss is defined as follows:

$$\mathcal{L}_{\text{feat}} = \frac{1}{M}\sum_{j=1}^{M}\sum_{(l,l')\in\mathcal{P}}\frac{\alpha_{l,l'}}{C_{l'}H_{l'}W_{l'}}\cdot\left\|T\left(D_p^{(l)}(x_j^d)\right) - D_{MFI}^{(l')}(x_j^d)\right\|_2^2 \tag{2}$$

Here, $\mathcal{P}$ denotes the set of layer-wise distillation pairs between the prior Stable Diffusion decoder D_p and the MFI decoder D_{MFI} (e.g., $\mathcal{P} = \{(1,4),(2,3),(3,2)\}$), where features from earlier layers of the SDI decoder are aligned with deeper layers of the MFI decoder. $T(\cdot)$ is an optional transformation function (which might be a 1×1 convolution or a multilayer perceptron) to ensure feature compatibility between the paired layers. The coefficient $\alpha_{l,l'}$ adjusts the contribution of each distillation pair, and $\|\cdot\|_2^2$ denotes the squared L2 loss. This feature-level supervision is designed to encourage the MFI model to reconstruct more semantically consistent and structurally accurate outputs by mimicking the informative hierarchical representations decoded by M_p.

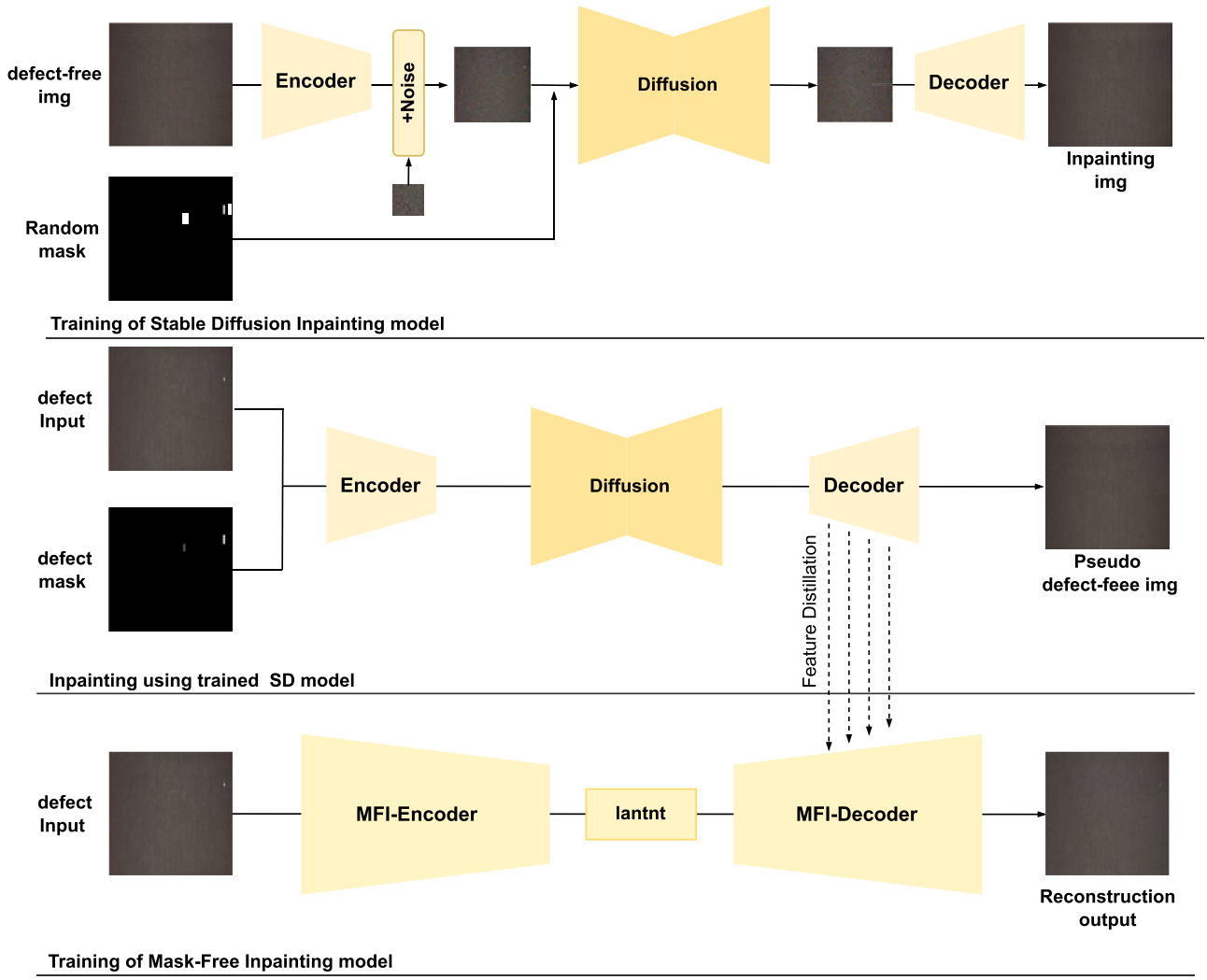

Fig. 2. Mask-Free Inpainting (MFI). (1) Training of Stable Diffusion Inpainting shows the training of SDI using defect-free images and random masks; (2) Inpainting using a trained SD model generates defect-free images from defect images and their defect masks; (3) Training of Mask-Free Inpainting model demonstrates the training of MFI, which distills features to its decoder from the SD decoder. Additionally, MFI uses the pseudo defect-free images as supervisory targets to generate reconstructed output.

In addition to the feature-level distillation loss, we introduce a pixel-wise reconstruction loss to supervise the output quality of the MFI module. This loss is defined for both defect-free and defective images, ensuring that the MFI model consistently generates defect-free results regardless of the input image condition. We employ the reconstructed defective image dataset $\hat{\mathcal{D}}_d$ as the supervision targets for the defect image dataset $\mathcal{D}_d$ to guide the MFI model in removing defects rather than merely copying the input. The overall pixel-level loss is formulated as:

$$\mathcal{L}_{\text{pix}} = \frac{1}{N}\sum_{i=1}^{N}\left\|\hat{x}_{df}^{n} - x_i^n\right\|_1 + \frac{1}{M}\sum_{j=1}^{M}\left\|\hat{x}_j^d - \hat{x}_j^{\text{M}}\right\|_1$$

where $\hat{x}_{df}^n$ and $\hat{x}_j^d$ denote the outputs of the MFI model for defect-free and defective inputs, respectively, and x_j^{M} denotes the SD inpainting of x_j^d. The first term enforces identity reconstruction for defect-free inputs, while the second term encourages the MFI model to learn defect inpainting behavior for corrupted inputs.

The final objective for optimizing the MFI module is defined as:

$$L_{MFI} = \lambda_{pix} L_{pix} + \lambda_{feat} L_{feat} \tag{3}$$

where λ_{pix} and λ_{feat} is the loss weights of $L_{pix}$and L_{feat}, respectively.

In summary, MFI reconstructs defect-free fabric textures without requiring prior defect masks, enabling reliable residual-based cues that serve as the foundation for discovering unknown defects in the open-world setting.

3.2. Dual-view pseudo-label generator (DVPG)

To generate reliable pseudo-labels for unknown classes, we propose the Dual-View Pseudo-label Generator (DVPG) module, which leverages an image pair consisting of a defective image and its corresponding reconstruction produced by the MFI module to generate fine-grained pseudo-labels. Given a defective input image x_d and its corresponding reconstructed image $\hat{x}_{re}$, we define the forward propagation of DVPG as follows:

$$\left\{\mathcal{F}(x_d), \mathcal{F}(\hat{x}_{re})\right\} \tag{4}$$

where $\{\mathcal{F}(\cdot)\}$ denotes a shared-weight encoder network that extracts semantic feature maps. In our study, ground truth is a binary image, where defect pixels are labeled as 1. The convolutional kernels in DVPG are designed to compute pixel-wise residuals, minimizing output values in defect-free regions and maximizing them in defective ones, thereby improving defect detection. As a result, defect pixels are projected to the value of 1, ensuring clear differentiation between defective and non-defective areas. Additionally, DVPG further computes feature-level differences to highlight defect regions more distinctly. A decoder module

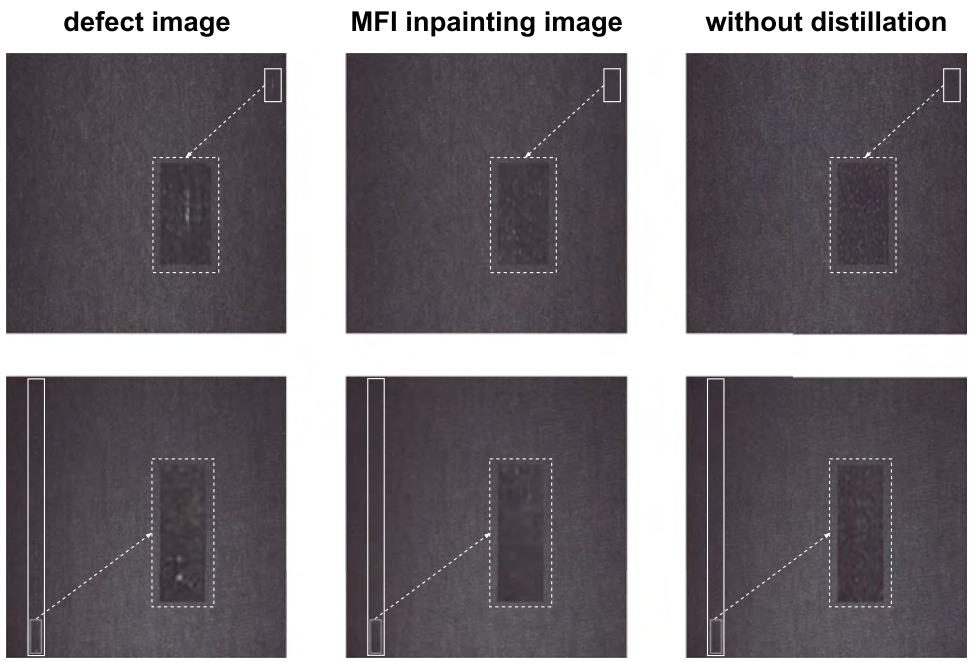

Fig. 3. Qualitative reconstruction results of the Mask-Free Inpainting (MFI) module with and without feature-level distillation. The upper row shows reconstructions produced by the proposed MFI model with feature distillation from the prior diffusion decoder, which yields smoother texture continuity and fewer residual defect-like artifacts. The lower row corresponds to reconstructions generated without feature distillation, where incomplete texture recovery and structural inconsistencies can be observed. Each column represents a different fabric sample.

processes these differences to generate the pixel-wise defect segmentation output, $\hat{m} \in [0,1]^{H\times W}$, representing the probability of each pixel being defective.

To optimize DVPG for accurate and structurally coherent segmentation, we employ a joint loss consisting of: Binary Cross-Entropy (BCE) loss for pixel-wise classification, ensuring segmentation accuracy:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[m_i \log \hat{m}_i + (1-m_i)\log(1-\hat{m}_i)\right] \tag{5}$$

Dice loss to enhance structural coherence in defect regions and suppress isolated fragments:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum m_i \hat{m}_i}{\sum m_i + \sum \hat{m}_i} \tag{6}$$

The final loss function is:

$$\mathcal{L}_{\text{DVPG}} = \lambda_1 \mathcal{L}_{\text{BCE}} + \lambda_2 \mathcal{L}_{\text{Dice}} \tag{7}$$

where m denotes the ground-truth defect mask, $\hat{m}$ represents the predicted mask region, and λ_1, λ_2 are hyperparameters weighting the loss components.

Given the segmentation output $\hat{m}$ produced by DVPG, our objective is to transform this continuous-valued mask into a set of bounding boxes that can serve as supervisory signals for the detection of unknown classes. To this end, we design a unified region localization pipeline that takes $\hat{m}$ as input and progressively refines it into high-quality bounding box annotations. Specifically, we apply a threshold function $T = 0.5$ to obtain a binary map $\hat{m}_{\text{bin}}$, where pixels with confidence scores exceeding the threshold are marked as potential defect regions, thereby suppressing background noise.

Following this, connected component analysis is conducted on $\hat{m}_{\text{bin}}$ to identify spatially contiguous foreground regions. Each connected component is then enclosed by its minimal axis-aligned bounding box $B_i = x_i, y_i, w_i, h_i$, where (x_i, y_i) denotes the top-left corner and (w_i, h_i) the width and height of the box, respectively.

To mitigate the effects of noise and over-segmentation, we implement a size filter by setting minimum width and height constraints $(w_{\min}, h_{\min})$; components failing to meet these criteria are excluded as unreliable. Furthermore, redundant or overlapping boxes are consolidated through Non-Maximum Suppression (NMS), using an IoU-based criterion to select the most confident proposal in overlapping regions.

$$\text{IoU}(B_i, B_j) = \frac{|B_i \cap B_j|}{|B_i \cup B_j|} \tag{8}$$

The resulting set of refined bounding boxes $\mathcal{B}_{\text{final}}$ provides precise localization cues and serves as high-quality supervision for training the subsequent defect detection module, as shown in Fig. 4.

Overall, DVPG leverages both pixel-level residuals and feature-level discrepancies to generate reliable pseudo-labels for unknown defects, which is critical for supervising incremental learning when ground-truth annotations are unavailable.

3.3. Decoupled incremental learning (DIL)

To generate reliable pseudo-labels and effectively decouple the learning process of known and unknown defect classes, we propose a label filtering strategy, as shown in Fig. 5. Specifically, a YOLO11 detector trained on the known-class dataset $\mathcal{D}_{known}$ is employed to predict a dense set of bounding boxes $\mathcal{B}_c$ on the defect image dataset $\mathcal{D}_d$ by relaxing the confidence threshold to $\tau = 0.01$. This approach allows the detector to output a large number of low-confidence detections, ensuring broader coverage of potential defect regions.

Given the set of candidate bounding boxes $\mathcal{B}_{\text{final}}$ provided by DVPG, each region $b_i \in \mathcal{B}_{final}$ is evaluated based on its spatial overlap with the YOLO predictions $\mathcal{B}_c$. An Intersection over Union (IoU) computation is performed between each b_i and all boxes b_i^c in $\mathcal{B}_c$. If b_i aligns with any b_i^c such that the IoU exceeds a predefined threshold γ and the matched prediction possesses a high confidence score s, then it is classified as a known-class instance and is excluded from the pseudo-label pool. Conversely, if there is insufficient overlap or if the matched predictions exhibit a low confidence score s, then the candidate box b_i is designated as an unknown-class instance and is included in the unknown-class dataset $\mathcal{D}_{unknown}$ for subsequent learning.

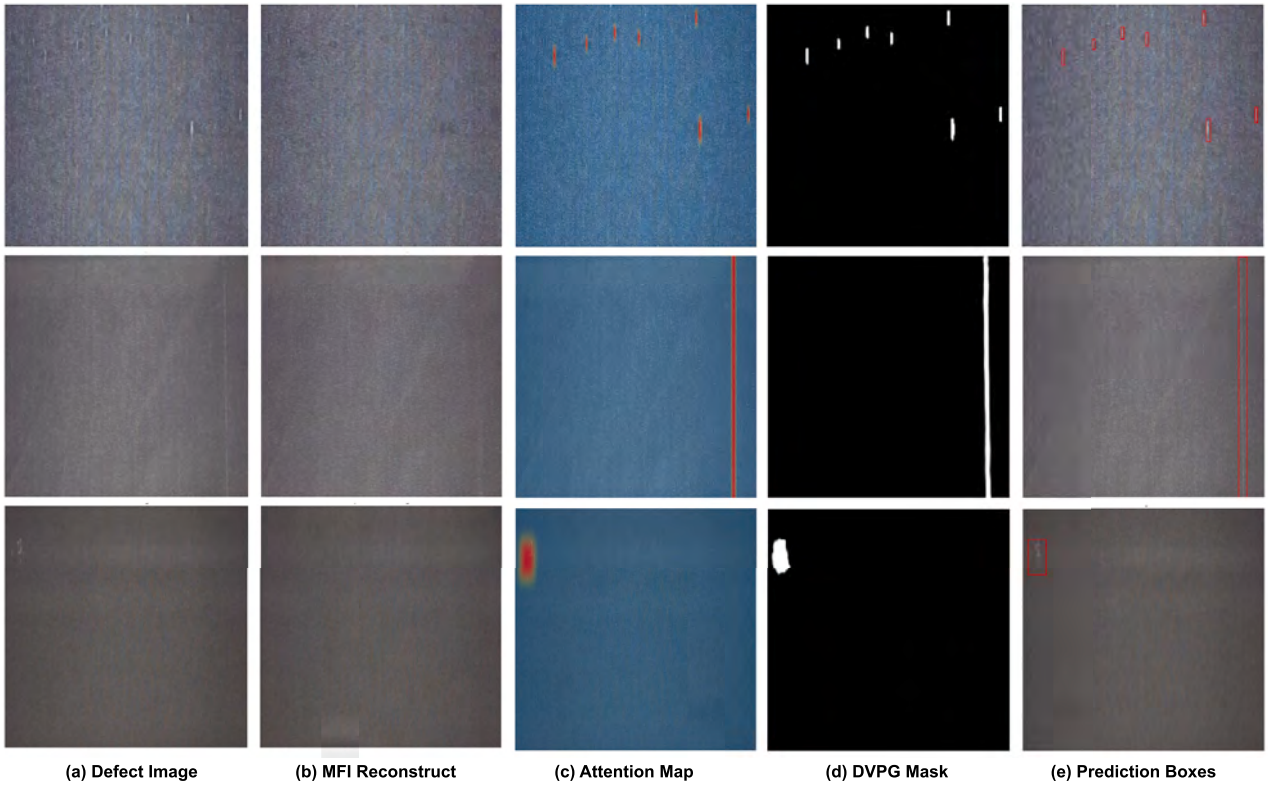

Fig. 4. Qualitative visualization of the defect localization process across OW-DLN stages. Each row corresponds to one complete example, showing the sequential outputs of all modules in our framework. (a) **Defect Image** represents the original fabric image containing visible or subtle defects. (b) **MFI Reconstruct** denotes the defect-free image generated by the Mask-Free Inpainting module. (c) **Attention Map** visualizes the defect-guided spatial attention produced by DAAM (CBAM-style channel→spatial attention with defect guidance), highlighting regions where the network focuses on potential defect areas. (d) **DVPG Mask** shows the binary pseudo-labels generated by the Dual-View Pseudo-segmentation module. (e) **Prediction Boxes** indicate the final defect detection results obtained after post-processing. The results demonstrate that DAAM effectively sharpens the localization of small or low-contrast defects by directing attention toward defect-prone texture regions.

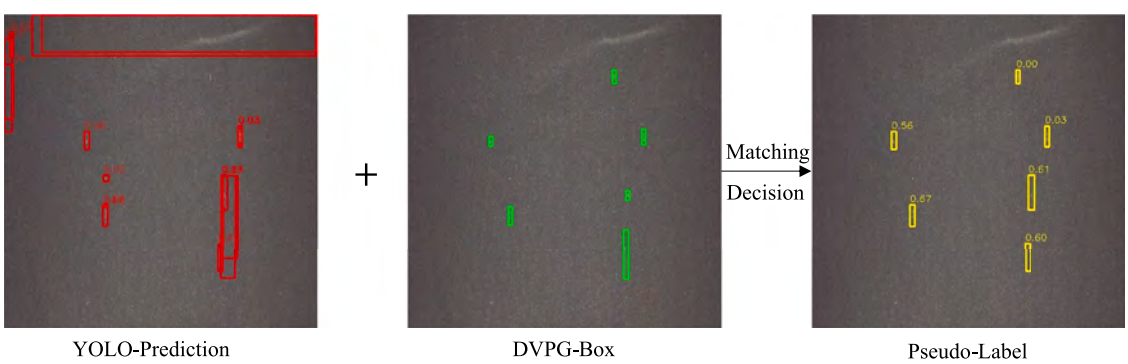

Fig. 5. Illustration of the pseudo-label filtering process. The proposed filtering rule integrates DVPG-generated pseudo boxes with low-confidence YOLO detector predictions ($\tau = 0.01$). For each pseudo box, the detector box with the highest IoU is found. If the matched detector box satisfies IoU$\geq \gamma$ and Score$Score \geq s$, it is considered a known sample and removed; otherwise, it is retained as an unknown pseudo label. The resulting cleaner pseudo-label pool provides reliable unknown defect candidates for subsequent incremental learning.

Addressing the challenge of catastrophic forgetting during model updates, we propose a Parallel Learning Network (PLN), which segregates the feature learning of known and unknown classes, as illustrated in Fig. 6. The proposed Decoupled Incremental Learning Network (DIL) utilizes a dual-path architecture that integrates a frozen known-class backbone with a trainable unknown-class branch. To ensure stable gradient propagation and prevent feature interference between the two paths, we insert zero-initialized 1×1 convolution layers at the connection points between the frozen and trainable branches. These layers serve as learnable residual adapters, enabling the unknown branch to incrementally assimilate novel knowledge without disturbing the pre-trained representations of the known branch. This approach is inspired by the ControlNet architecture [24], which demonstrates that zero-initialized side connections can effectively stabilize training during the extension of pre-trained diffusion backbones.

The PLN effectively decouples the learning process by routing D_{known} and D_{unknown} through distinct branches. During joint training, feature fusion is limited to shallow levels to maintain low-level consistency, while deeper features are kept independent.

Furthermore, we integrate a Defect-Aware Attention Module (DAAM), as shown in Fig. 7, to enhance the localization of defect regions for both known and unknown classes. Practically, DAAM is embedded into three feature extraction stages of the YOLO11 backbone–after the 3rd, 4th, and 5th convolutional blocks, which correspond to the feature

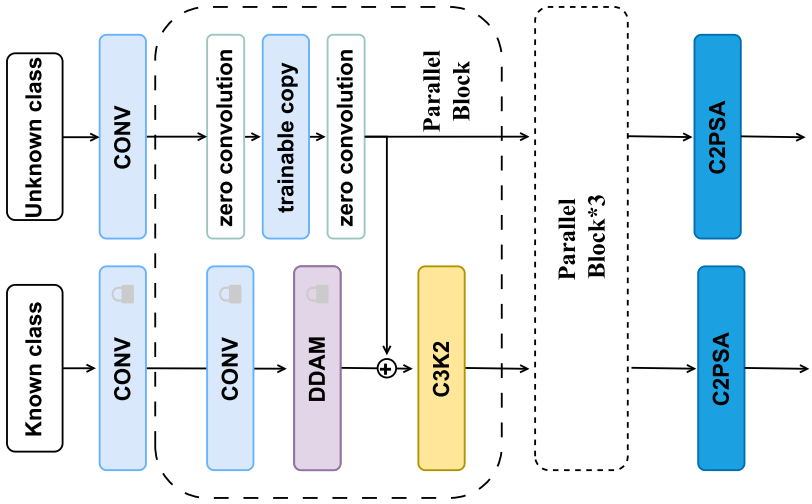

Fig. 6. Decoupled Incremental Learning Network. The left side maintains a locked (non-trainable) known-class learning backbone on known classes, which during training incorporates an unknown-class learning backbone copy branch on the right side to learn novel categories while mitigating catastrophic forgetting in the model.

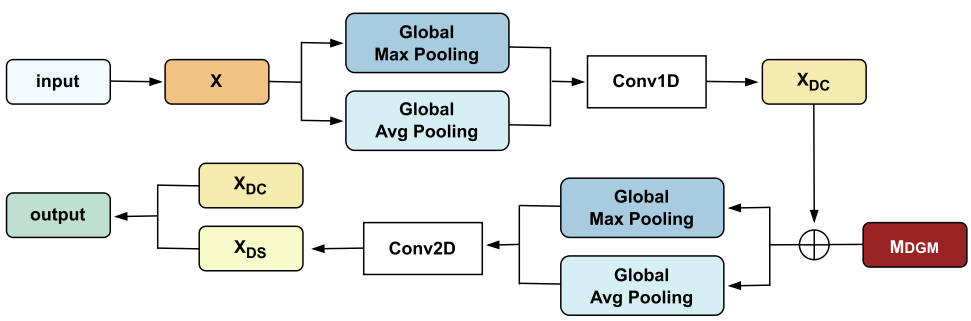

Fig. 7. The illustration of attention module.

maps commonly referred to as P3, P4, and P5 (with strides of 8, 16, and 32, respectively). In simpler terms, these stages represent the network's progressively deeper feature layers: P3 focuses on fine texture patterns, P4 captures structural details, and P5 encodes high-level semantic information. By inserting DAAM at these three scales, we refine both local texture cues and global semantic representations before passing them to the neck and detection head for final prediction.

Firstly, DAAM adopts a lightweight structure that utilizes 1D convolutions rather than MLPs to compute inter-channel dependencies,thereby reducing computational costs. To improve defect perception, features are extracted through Global Average Pooling and Global Max Pooling, with channel weights subsequently computed via 1D convolutions:

$$\mathbf{F}_{\text{avg}}(c) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} \mathbf{X}_{c,i,j} \tag{9}$$

$$\mathbf{F}_{\max}(c) = \max_{1 \le i \le H} \max_{1 \le j \le W} \mathbf{X}_{c,i,j} \tag{10}$$

$$\mathbf{F}_{\text{ch}} = \sigma\big(\text{Conv1D}\big([\mathbf{F}_{\text{avg}}, \mathbf{F}_{\max}]\big)\big) \tag{11}$$

where $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$ is the input feature map with height H and width W, Conv1D represents the 1D convolution operation, and σ is the sigmoid activation function. The channel-refined features are derived as follows:

$$\mathbf{X}_{DC} = \mathbf{X} \cdot \mathbf{F}_{ch} \tag{12}$$

Secondly, DAAM incorporates a defect-guided mask $\mathbf{M}_{\text{DGM}}$ which is derived from the outputs of DVPG to enhance the spatial focus on defect-prone areas. Initially, spatial features are computed through global max-pooling and average-pooling:

$$\mathbf{F}_{\text{sp}} = \sigma\big(\text{Conv2D}\big(\big[\mathbf{F}_{\max}(\mathbf{X}_{\text{DC}}), \mathbf{F}_{\text{avg}}(\mathbf{X}_{\text{DC}})\big]\big)\big) \tag{13}$$

$$\mathbf{F}_{\max}(i,j) = \max_{c \in C} \mathbf{X}_{\text{DC}}(c,i,j), \tag{14}$$

$$\mathbf{F}_{\text{avg}}(i,j) = \frac{1}{C} \sum_{c=1}^{C} \mathbf{X}_{\text{DC}}(c,i,j) \tag{15}$$

The spatial attention is further enhanced by incorporating the defect-guided mask M_{DGM}

$$\mathbf{X}_{\text{DS}} = \mathbf{X}_{\text{DC}} \cdot \big(\mathbf{F}_{\text{spatial}} + \lambda \mathbf{M}_{\text{DGM}}\big) \tag{16}$$

where λ is a tunable parameter balancing the influence of defect-aware guidance.

This enhanced attention representation strengthens the model's sensitivity to defect-related features across both branches, further improving detection accuracy in open-world and incremental scenarios, as shown in Fig. 4.

Formally, let F_k and F_u denote the feature maps from the known and unknown branches, respectively. The final fused feature is computed as:

$$F = \alpha F_k + (1-\alpha) F_u, \quad \alpha \in [0,1] \tag{17}$$

where α can balance the influence of the known and unknown feature paths.

To ensure representation consistency and prevent feature drift during joint learning, an alignment loss is introduced:

$$\mathcal{L}_{\text{align}} = \|F_k - \text{StopGrad}(F_u)\|_2^2 \tag{18}$$

Table 1
Category statistics of the ZJU-Leaper dataset.

Category	Box Count	Image Count	Boxes/Image
Dot Pattern	20,598	13,792	1.49
Thick Stripe	1366	684	2.00
Thin Stripe	974	677	1.44
White Plain	845	608	1.39

Table 2
Comprehensive Parameter Settings.

Parameter	Value	Parameter	Value
Batch size	8	Epoch	300
Image size	1504	Learning rate	0.01
Momentum	0.937	Weight decay	0.0005
MFI λ_{pix}	1	MFI λ_{feat}	0.5
DVPG λ_1	1	DVPG λ_2	1
DIL λ (align)	0.5	DAAM λ	0.1
MFI Optimizer	Adam	MFI LR	0.0002
MFI Batch size	16	MFI Epochs	200
Binarization threshold T	0.5	Min box size	(16,16)
Score threshold τ	0.01	IoU threshold γ	0.5

This loss promotes low-level consistency between known and unknown features while blocking gradient flow from the unknown to the known branch, thereby preserving previously acquired knowledge and mitigating catastrophic forgetting.

The overall DIL loss is formulated into:

$$\mathcal{L}_{\text{DIL}} = \mathcal{L}_{\text{det}}^{\text{known}} + \mathcal{L}_{\text{det}}^{\text{unknown}} + \lambda \cdot \mathcal{L}_{\text{align}} \tag{19}$$

where $\mathcal{L}_{\text{det}}$ is the standard detection loss (classification + localization).

In essence, DIL decouples the representation learning of known and unknown defects, allowing the detector to incrementally absorb new defect categories while effectively mitigating catastrophic forgetting. DAAM further enhances OW-DLN by adaptively emphasizing defect-relevant regions in feature maps, improving both localization accuracy and discrimination between known and unknown defects under complex fabric textures.

4. Experiments

4.1. Datasets

We evaluate OW-DLN on the Tianchi Textile Defect Dataset [25], which contains 9523 images spanning 34 defect categories. Following the official guidelines [25], defect classes with similar causes or visual patterns are merged, resulting in 20 unified classes, as shown in Fig. 8. The dataset and detailed descriptions are publicly available on the Tianchi website[1]. To realistically simulate industrial open-world scenarios–where only a subset of defect types is initially labeled and new defects emerge over time–we divide the dataset into two tasks. Considering the severe class imbalance, we construct only two tasks: defect classes with relatively abundant samples are selected as known classes in Task 1, while less frequent classes are introduced later in Task 2. Task 1 (Open-world Defect Detection): We train the model using 5 known defect classes (wrapping knot, float, hole, mispick, and starch lump), while both known and unknown classes appear during testing. This task evaluates the model's ability to detect unknown defects without prior exposure. Task 2 (Incremental Learning): Based on Task 1, the remaining 15 defect classes are gradually introduced through incremental learning. This task assesses the model's capability to learn new defect categories while retaining performance on previously learned classes, thereby mitigating catastrophic forgetting.

[1] https://tianchi.aliyun.com/dataset/79336

In addition to the Tianchi dataset, we further evaluate the OW-DLN on the ZJU-Leaper dataset [26], a public benchmark for fabric defect detection developed by Zhejiang University. This dataset includes multiple fabric styles and four representative defect categories classified according to their structural or visual characteristics: Dot Pattern, Thick Stripe, Thin Stripe, and White Plain. These categories respectively represent local point-like flaws, wide structural stripe defects, narrow linear texture anomalies, and brightness inconsistency or density variation regions, covering typical geometric patterns observed in industrial fabric inspection.

To align with the open-world and incremental learning settings, we use the Dot Pattern class, which has the largest number of annotated samples, as the known category in Task 1, while the remaining three categories (Thick Stripe, Thin Stripe, and White Plain) are treated as unknown classes during testing. In Task 2, all four categories are incrementally introduced to evaluate the model's ability to learn new defect types without catastrophic forgetting. The distribution of these categories is summarized in Table 1.

To further evaluate our model under real-world conditions, we have amassed a validation set of 1605 knitted fabric defect images using a multi-view image acquisition system. These images encompass ten typical defect types, including snag, knot, pulled yarn, hole, color yarn, yarn nep, oil stain, dirt, foreign yarn, and writing. We split it into two tasks as the Tianchi dataset: Task 1 (Open-world Defect Detection): In this task, we use 5 defect classes (Snag, Knot, Pulled yarn, Hole and Color yarn) for training. Task 2 (Incremental Learning): Building on Task 1, the 5 unknown classes (Oil stain, Dirt, Yarn nep, Foreign yarn and Writing) are gradually introduced for training through an incremental learning process.

4.2. Implementation details and evaluation metrics

Our OW-DLN is implemented using the PyTorch framework with Python, utilizing an Nvidia GeForce RTX A5000 24G GPU. The hyperparameters are delineated in Table 2.

To quantitatively evaluate the proposed OW-DLN from the perspectives of unknown discovery, class discrimination, and knowledge retention, we adopt the following metrics: U-Recall [4]: measures the model's ability to discover unknown defects during inference, defined as the recall of objects labeled as "unknown". WI (Wilderness Impact) [4]: quantifies the confusion between known and unknown categories by evaluating the degradation of performance in knownclasses when unknown samples are introduced. A-OSE (Absolute Open-Set Error) [4]: counts the number of unknown objects mistakenly classified as known classes, reflecting the model's robustness to unknown interference. mAP@50 [27]: denotes the mean Average Precision at an IoU threshold of 0.5, used to evaluate the retention of prior knowledge and detection accuracy for both previously and currently known classes.

To ensure the fairness of comparison, all methods are trained and evaluated under identical experimental conditions, including dataset splits, image resolutions, training epochs, and optimization strategies.

4.3. Comparison with state-of-the-art methods

4.3.1. Open-world defect detection

We evaluate the proposed OW-DLN against state-of-the-art OWOD methods on two benchmarks, Tianchi Textile Defect and ZJU-Leaper; the evaluation results are presented in (Fig. 9) and (Fig. 10). The evaluation encompasses: (1) detecting both known and unknown defects (Task 1), and (2) incremental learning under continual category expansion (Task 2). We average all results over three random seeds (0, 42, 2025) and report them as mean ± standard deviation for statistical reliability. As summarized in Tables 3 and 4, OW-DLN consistently outperforms existing OWOD frameworks across all metrics and datasets, verifying its robustness and generalization in diverse textile domains.

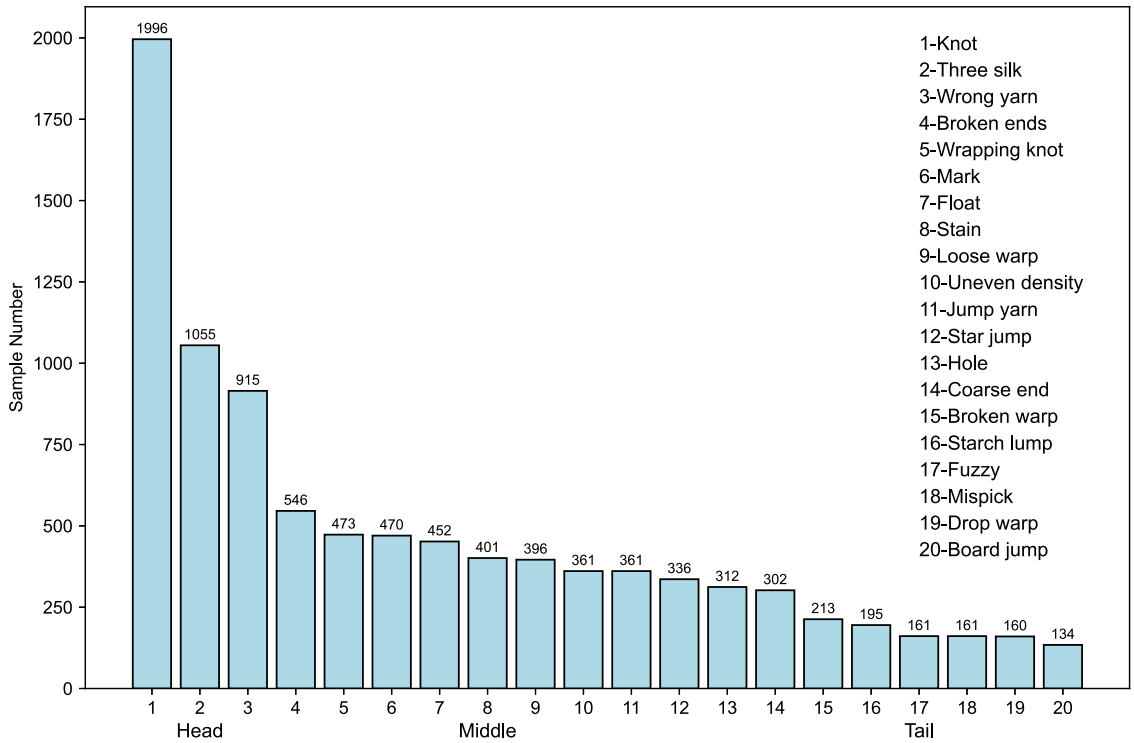

Fig. 8. Class distribution of Tianchi Textile Defect Dataset after merging. We divide the data into 3 parts: head, middle, and tail according to the number of samples of each class. We select 1 class (wrapping knot) from the head, 2 classes (float and hole) from the middle, and 2 classes (mispick and starch lump) from the tail as known classes for Task 1. The remaining 15 classes are used as unknown classes for Task 2.

Table 3
Performance comparison of Open-world fabric defect detection on Tianchi Textile Defect Dataset.

Model	Task 1				Task 2		
	U-Recall (↑)	WI (↓)	A-OSE (↓)	mAP@50 (Current known) (↑)	mAP@50 (Prev. known) (↑)	Current known (↑)	Both (↑)
ORE [4]	0.035	0.128	312	0.342	0.215	0.146	0.163
OW-DETR [19]	0.055	0.110	276	0.403	0.276	0.163	0.214
PROB [28]	0.072	0.102	251	0.487	0.342	0.219	0.291
RandBox [29]	0.064	0.098	247	0.441	0.312	0.203	0.254
Hyp-OW [27]	0.085	0.096	241	0.528	0.431	0.288	0.315
YOLO-World [20]	0.102	0.090	214	0.324	0.216	0.256	0.345
OW-DLN (Ours)	**0.153±0.005**	**0.072±0.005**	**148±5**	**0.680±0.009**	**0.602±0.008**	**0.460±0.010**	**0.502±0.011**

Table 4
Performance comparison of Open-world fabric defect detection on ZJU-Leaper dataset.

Model	Task 1				Task 2		
	U-Recall (↑)	WI (↓)	A-OSE (↓)	mAP@50 (Current known) (↑)	mAP@50 (Prev. known) (↑)	Current known (↑)	Both (↑)
ORE [4]	0.041	0.132	260	0.382	0.244	0.165	0.188
OW-DETR [19]	0.058	0.118	157	0.446	0.296	0.204	0.236
PROB [28]	0.076	0.107	127	0.509	0.358	0.249	0.297
RandBox [29]	0.069	0.103	103	0.474	0.333	0.229	0.271
Hyp-OW [27]	0.092	0.096	99	0.548	0.441	0.308	0.334
OW-DLN (Ours)	**0.302±0.007**	**0.072±0.002**	**85±3**	**0.736±0.0009**	**0.708±0.006**	**0.782±0.006**	**0.717±0.005**

On the Tianchi dataset, the OW-DLN achieves the highest scores across all evaluated metrics. It attains a U-Recall of 0.153, exceeding the best baseline Hyp-OW, and a mAP@50 of 0.680, which confirms its superior ability in identifying unknown defects while preserving accuracy on known defects. Additionally, the model achieves the highest WI and A-OSE scores, reflecting a balanced precision-recall trade-off and strong robustness under open-world conditions.

Qualitative analysis reveals that YOLO-World tends to produce overgeneralized predictions, labeling previously unseen defects (e.g., yarn neps or oil stains) as semantically similar known types (e.g., "spot" or "dot"). This behavior stems from its visual-language pretraining paradigm, which prioritizes high-level semantic alignment rather than fine-grained texture discrimination. Consequently, YOLO-World exhibits strong zero-shot transferability on generic objects but fails to capture subtle local irregularities that are crucial for textile defect recognition. These findings indicate that open-vocabulary detection (OVD) frameworks are not directly applicable to open-world defect detection tasks without domain-specific adaptation for industrial textures and mechanisms for continual category updates.

In Task 2 (incremental learning), the OW-DLN achieves scores of 0.602 on previously known categories and 0.460 on newly introduced categories, yielding a top combined score of 0.502. Compared with

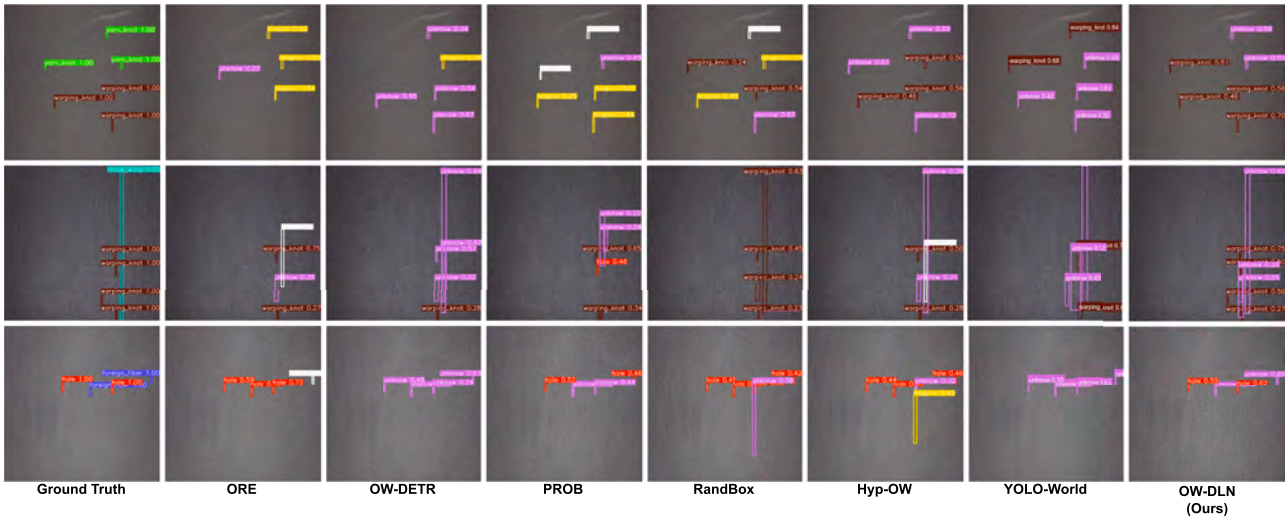

Fig. 9. State-of-the-art comparison on the Open-world fabric defect detection task on Tianchi Textile Defect Dataset.

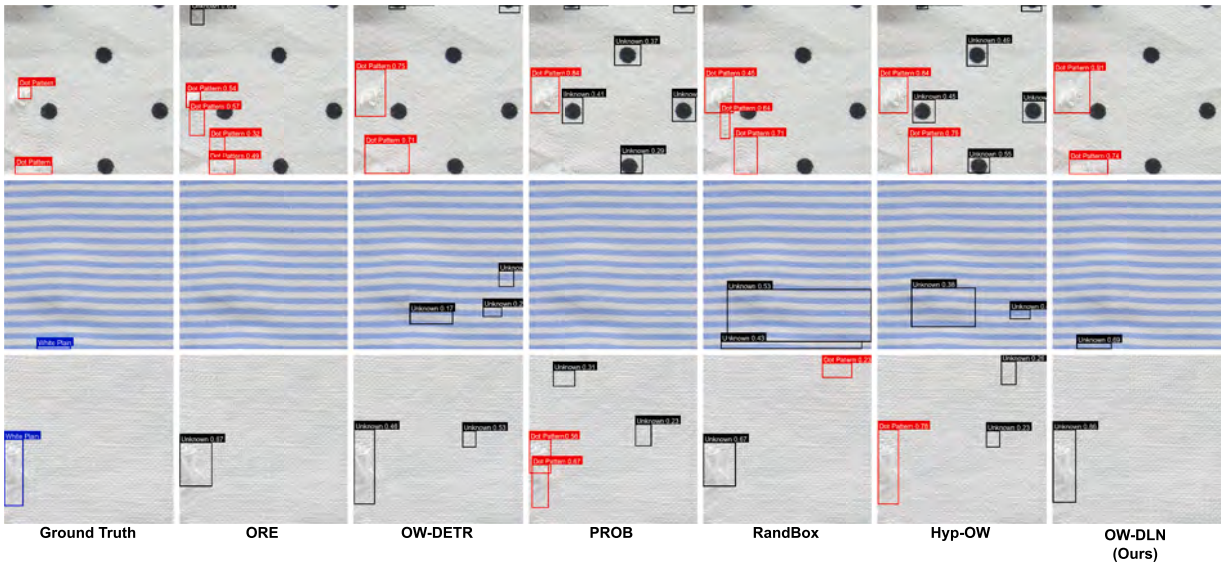

Fig. 10. State-of-the-art comparison on the Open-world fabric defect detection task on ZJU-Leaper Dataset.

Hyp-OW (0.431/0.288/0.315), it improves by 39.7%, 59.7%, and 59.4% on the respective metrics, indicating effective knowledge retention and fast adaptation during continual learning. Unlike prior methods, which suffer severe performance degradation after updates, OW-DLN maintains stable performance across evolving categories.

Results on ZJU-Leaper further validate the generalization of OW-DLN to fabrics characterized by complex textures and fine-grained boundaries. It achieves a U-Recall of 0.302, over three times higher than that of Hyp-OW, and maintains high known-class accuracy (mAP@50 = 0.736). In Task 2, OW-DLN once again delivers superior performance (0.708 for previous and 0.782 for current classes), outperforming the best baseline. These results confirm that OW-DLN effectively preserves prior knowledge while rapidly adapting to new defect types.

Overall, the cross-dataset results substantiate that OW-DLN generalizes effectively to varying textile materials and imaging conditions. The superior performance can be attributed to the synergy of its three key components: the Mask-Free Inpainting (MFI) module that reconstructs defect-free regions to aid in the discovery of unknown classes, the Dual-View Pseudo-label Generator (DVPG) that provides reliable supervision for novel categories, and the Decoupled Incremental Learning (DIL) strategy that mitigates catastrophic forgetting. Together, these components enable OW-DLN to achieve stable and high-accuracy performance in open-world defect detection across both benchmark datasets.

Table 5
Comparison of different methods.

Methods	mAP	mAP@50
RetinaNet [30]	0.088	0.235
Faster RCNN [31]	0.132	0.320
PAFPN [32]	0.147	0.336
Mask RCNN [33]	0.130	0.307
Libra RCNN [34]	0.116	0.283
Ms RCNN [35]	0.132	0.305
DCN [36]	0.139	0.330
Swin-Transformer [37]	0.154	0.346
JDCBL [38]	0.170	0.393
OW-DLN	**0.230±0.008**	**0.510±0.012**

4.3.2. *Fabric defect detection*

To further validate the efficacy of OW-DLN within standard (closed-world) detection settings, this study conducts comparison with leading fabric defect detection models. As depicted in Table 5 and Fig. 11,

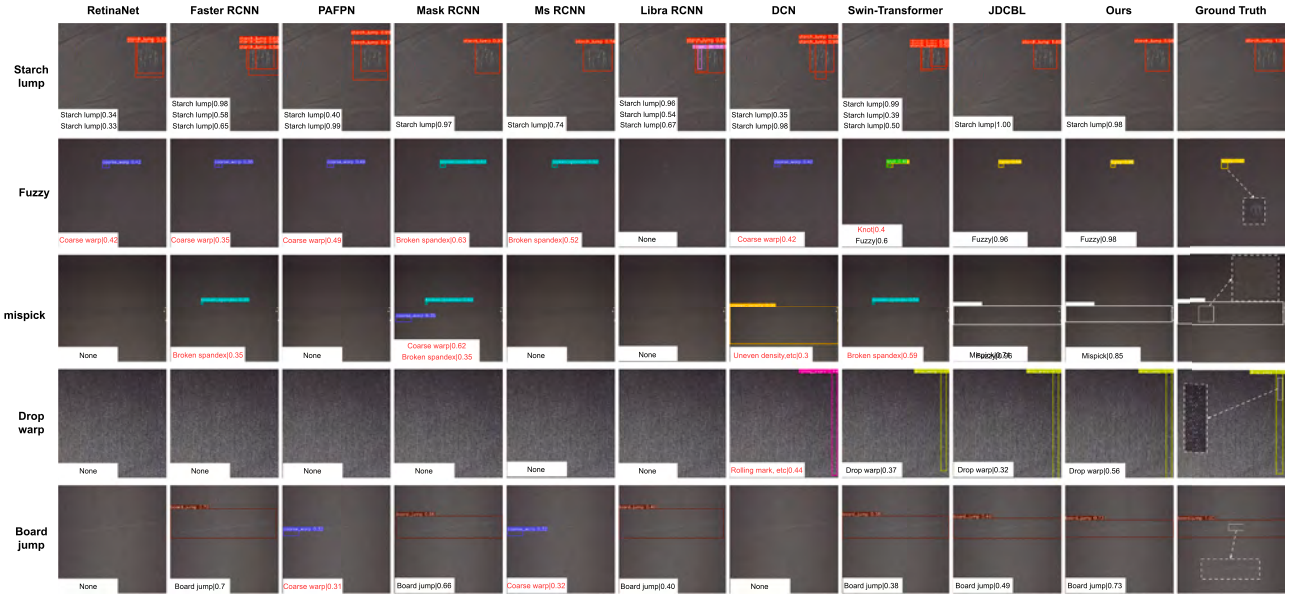

Fig. 11. Comparison of different methods in some tail classes.

Table 6
Computational complexity and inference efficiency at 512×512 input.

Model	Params (M)	GFLOPs	FPS (↑)	Remarks
ORE [4]	44.2	76.5	34.1	Faster R-CNN backbone
OW-DETR [19]	42.5	93.3	29.2	Transformer-based DETR
PROB [28]	49.1	80.9	36.1	Probabilistic branch
RandBox [29]	28.3	58.7	66.1	YOLO-like one-stage
Hyp-OW [27]	33.1	65.8	57.5	Hyperbolic embedding model
OW-DLN (Ours)	**20.5**	**48.2**	**86.3**	Inference uses backbone only (MFI/DVPG offline)

OW-DLN achieves a mAP of 0.230 and mAP@50 of 0.510, outperforming all baselines–including Faster RCNN [31], Mask RCNN [33], Swin-Transformer [37], and JDCBL [38].

These findings underscore that although primarily designed for open-world scenarios, OW-DLN also excels in conventional fabric defect detection tasks. The notable improvement in mAP@50 indicates that the model proficiently assimilates both global texture patterns and intricate local features, enabling accurate localization and classification across diverse defect types. This advantage stems from the synergy between its generative reconstruction (MFI) and decoupled learning (DIL) mechanisms, which enhance the feature representation of complex textures and strengthen robustness against intra-class variability.

Moreover, OW-DLN consistently upholds high detection quality across frequent and infrequent categories, showing stable convergence behavior and reduced overfitting compared with conventional CNN-based architectures. Its strong closed-world performance highlights the generality of the proposed framework–bridging the gap between open- and closed-world paradigms and highlighting its applicability for practical Implementation in industrial inspection systems where both known and unknown defects coexist.

4.3.3. Computational efficiency and model complexity

To evaluate the practical feasibility of OW-DLN in industrial inspection, we further compare its computational overhead and parameter count with representative OWOD baselines. All measurements are conducted on an NVIDIA RTX A5000 GPU with a batch size of 1 and input resolution of 512×512. Inference time is averaged over 500 images. As shown in Table 6, although OW-DLN incorporates a multi-stage design, its overall computational cost remains comparable to recent OWOD frameworks, while delivering significantly higher detection accuracy.

Although OW-DLN utilizes a multi-stage architecture, its *deployed* inference graph remains lightweight at 512×512: OW-DLN contains 20.5M parameters and requires 48.2 GFLOPs per image (Table 6), achieving approximately ~86.3 FPS (about 1.2 ms per image) on an RTX A5000. During real-time inference, only the trained detection backbone is executed–neither the MFI nor the DVPG modules are involved–so the runtime cost is close to modern one-stage detectors and only slightly slower than a base YOLO11s at the same resolution. Compared to transformer-based OW-DETR and two-stage ORE/PROB, OW-DLN delivers a markedly better accuracy-efficiency trade-off, fulfilling the real-time requirements of industrial applications.

4.4. Performance in wild-world dataset

To further probe OW-DLN's generalization capabilities in real-world scenarios, the model is tested on a wild-world fabric defect dataset containing both previously seen and newly introduced categories. Table 7 summarizes per-class Precision-Recall scores for Task 1 and Task 2, along with the corresponding mAP@50 outcomes.

Across both tasks, OW-DLN demonstrates robust adaptability and generalization in open-world conditions. It attains high Precision-Recall scores on typical defects such as hole (0.902), pulled yarn (0.856), and snag (0.892), while successfully identifying unseen anomalies with a 0.464 score for the unknown class in Task 1, yielding an overall mAP@50 of 0.782. After incremental training in Task 2, OW-DLN effectively incorporates new categories (oil stain, dirt, yarn nep, foreign yarn) while preserving accuracy on prior ones–e.g., knot improves from 0.670 to 0.843 and color yarn from 0.713 to 0.864. Newly added classes also perform well, with Precision-Recall scores of 0.762 (oil stain), 0.806 (dirt), and 0.842 (writing), achieving an overall mAP@50 of 0.753. These results confirm that OW-DLN maintains detection stability and

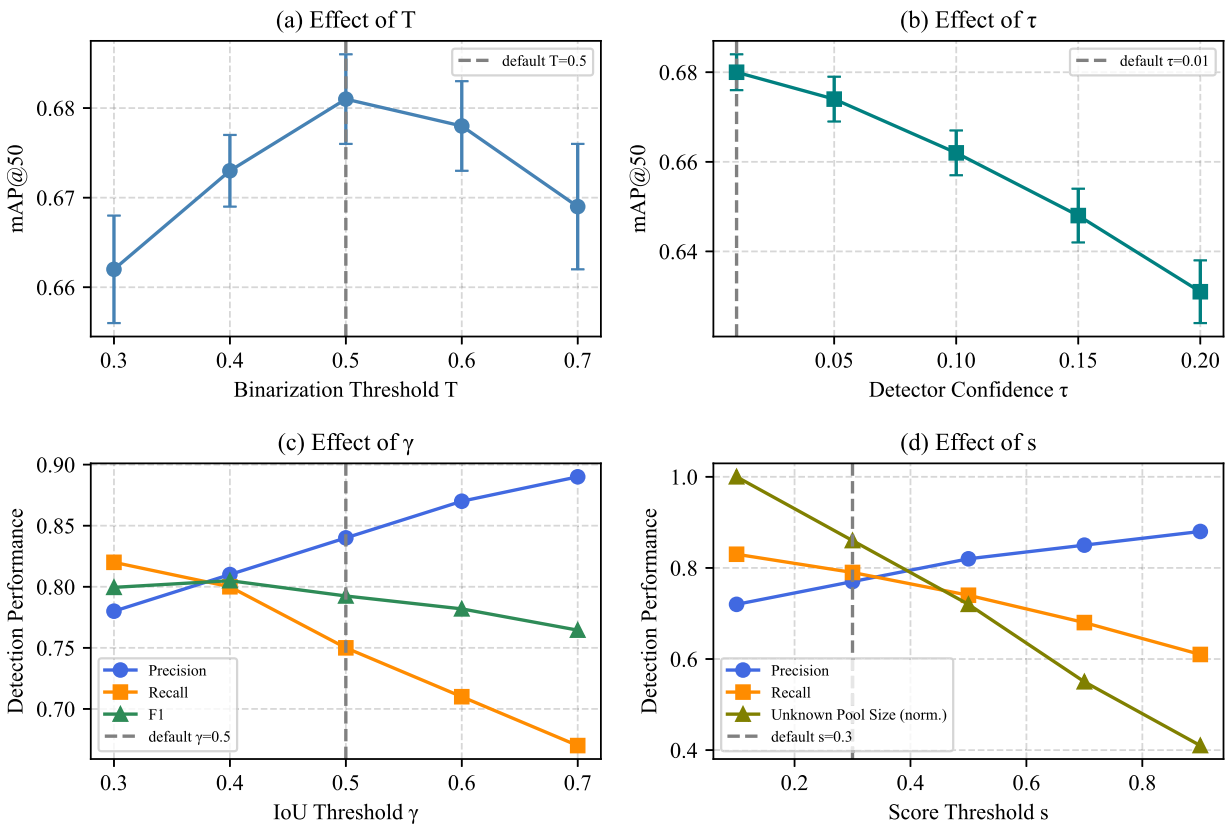

Fig. 12. Threshold-sensitivity analysis of OW-DLN. Each curve reports the mean ± standard deviation across multiple random seeds while varying one threshold and keeping the others fixed: (a) DVPG binarization threshold T; (b) detector confidence τ; (c) IoU gate γ; (d) detector score gate s. Plots (c) and (d) show Precision, Recall, and F1 for unk nowns, illustrating that stricter gates increase purity but may reduce recall. Vertical dashed lines mark the default configuration ($T = 0.5,\ \tau = 0.01,\ \gamma = 0.5,\ s = 0.3$).

Table 7
Per-class Precision–Recall results for Task 1 and Task 2.

Class	Task 1: Precision–Recall	Task 2: Precision–Recall
Snag	0.892	0.857
Knot	0.670	0.843
Pulled yarn	0.856	0.792
Hole	0.902	0.963
Color yarn	0.713	0.864
Oil stain	—	0.762
Dirt	—	0.806
Yarn nep	—	0.754
Foreign yarn	—	0.748
Writing	—	0.842
Unknown	0.464	—
mAP@50	**0.782**	**0.753**

knowledge retention across evolving categories evolve, establishing it as a scalable and reliable solution for open-world fabric defect inspection.

4.5. Ablation experiments

Table 8 presents the effect of progressively integrating our proposed components into the baseline framework for open-world fabric defect detection.

We initiate this study with a baseline YOLO11 detector without any open-world or incremental-learning mechanism. This configuration demonstrates moderate performance with a mAP@50 of 0.692 on the current known classes in Task 1. However, it exhibits severe catastrophic forgetting in Task 2, where the combined mAP@50 drops to only 0.315. This highlights that a closed-set detector cannot effectively adapt to new categories once the training distribution changes.

We integrate the DAAM into the C3–C5 layers of the backbone and the initial stage of neck fusion to enhance multi-scale texture perception. We evaluate three variants of this module: (i) channel-only, (ii) spatial-only, and (iii) the full version with defect-guidance. As shown in Table 8, incorporating either channel or spatial attention modestly improves the baseline performance (mAP@50: 0.692 to 0.721). However, the full version with defect-guidance provides the most significant boost, yielding mAP@50 scores of 0.730 for Task 1 and 0.332 for Task 2. This demonstrates that incorporating defect priors strengthens the model's response to localized features, enabling it to better detect subtle texture irregularities crucial for open-world defect detection.

Omitting the MFI module causes a sharp decline in performance, with the U-Recall dropping from 0.302 to 0.100, and the mAP@50 for Task 2 falling to 0.406. Similarly, removing the DVPG results in notable reductions: mAP@50 for Task 1 drops to 0.483, for Task 2 to 0.303, and U-Recall to 0.119. Without DVPG, the model struggles to differentiate genuine novel patterns from background noise. The full OW-DLN configuration achieves the best results across all metrics, with the highest U-Recall (0.302 ± 0.007), lowest WI (0.072 ± 0.002), and smallest A-OSE (85 ± 3). The mAP@50 scores of 0.736 for Task 1 and 0.717 for Task 2 confirm that the combination of MFI, DVPG, and the DIL mechanism within the DIL framework enables robust open-world generalization and stable continual adaptation.

4.6. Threshold sensitivity analysis

To verify the robustness of OW-DLN concerning the critical thresholds involved in pseudo-label filtering and reconstruction, a comprehensive parameter sensitivity analysis was conducted.

Table 8
Ablation study results.

Model	Task 1				Task 2		
	U-Recall (↑)	WI (↓)	A-OSE (↓)	mAP@50 (Current known) (↑)	mAP@50 (Prev. known) (↑)	Current known (↑)	Both (↑)
Baseline Detection Model	—	—	—	0.692	0.461	0.243	0.315
Baseline + DAAM (channel only)	0.108	0.089	196	0.705	0.478	0.258	0.321
Baseline + DAAM (spatial only)	0.115	0.085	184	0.721	0.482	0.267	0.326
Baseline + DAAM (+defect-guidance)	0.132	0.080	169	0.730	0.487	0.276	0.332
OW-DLN w/o MFI	0.100	0.091	212	0.580	0.513	0.405	0.406
OW-DLN w/o DVPG	0.119	0.083	178	0.483	0.317	0.292	0.303
OW-DLN (Ours)	**0.302±0.007**	**0.072±0.002**	**85±3**	**0.736±0.009**	**0.708±0.006**	**0.782±0.006**	**0.717±0.005**

The parameters evaluated encompass the DVPG binarization threshold T, the detector confidence threshold τ, the IoU matching gate γ, and the detector score gate s. Fig. 12 displays the sensitivity curves, described with mean ± SD across three random seeds, while maintaining all other experimental conditions constant.

As shown in Fig. 12(a) and (b), model performance remains stable across a wide range of T and τ, indicating that the proposed framework does not rely on delicate hyperparameter tuning. Fig. 12(c) and (d) further illustrate the effects of the two filtering thresholds γ and s on the validation set. Increasing γ from 0.3 to 0.7 reduces false keeps, improving Precision but slightly lowering Recall, while raising s from 0.1 to 0.9 progressively tightens the score gate, shrinking the unknown pseudo-label pool yet enhancing its purity. The F1 curve (green) confirms that both thresholds reach their optimal trade-off around $\gamma = 0.5$ and $s = 0.3$.

A vertical dashed line in each plot marks the default configuration ($T = 0.5,\ \ \tau = 0.01,\ \ \gamma = 0.5,\ \ s = 0.3$) used in all main experiments. The small standard deviations across seeds demonstrate the strong robustness and stability of OW-DLN under different parameter settings.

5. Conclusion

In this study, we propose OW-DLN, a novel open-world framework for fabric defect detection that effectively handles both known and unknown defect classes in dynamic industrial scenarios. OW-DLN achieves open-world defect detection by coupling generative texture reconstruction with defect-aware incremental recognition, allowing the system to reliably localize and expand to previously unseen defect types. By integrating latent-guide reconstruction, dual-view pseudo-label generator, and decoupled incremental learning, OW-DLN overcomes the limitations of traditional closed-set models and achieves superior performance in recognizing and adapting to new defect classes. Extensive experiments demonstrate that our model not only achieves state-of-the-art results on open-world detection tasks but also maintains strong competitiveness in conventional defect detection benchmarks. Ablation studies further confirm the effectiveness of each proposed module.

Despite the promising results, OW-DLN has certain limitations that also point to promising directions for future research. First, our experimental evaluation focuses primarily on fabric datasets. While these datasets encompass diverse texture patterns and defect characteristics, extending OW-DLN to other manufacturing domains–such as metallic surfaces, PCB inspection, or food quality control–requires further validation and adaptation. Second, OW-DLN leverages a pretrained generative prior distilled from a diffusion model to enhance defect-free reconstruction and pseudo-label quality. Although this design choice substantially improves performance within the textile domain, its effectiveness may be influenced by severe domain shifts, including extreme illumination changes, non-textile materials, or unseen imaging modalities. Incorporating domain-robust or self-adaptive generative priors represents a promising avenue to further strengthen the framework. Overall, these considerations do not detract from the effectiveness of OW-DLN in the fabric inspection setting, but rather highlight opportunities to broaden its robustness and applicability in real-world industrial inspection environments.

CRediT authorship contribution statement

Qingming Ye: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis; **Jue Hou:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Conceptualization; **Yang Yang:** Investigation, Data curation; **Jiahe Xue:** Visualization, Validation, Conceptualization; **Zheng Liu:** Supervision, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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